F78PA Probability & Statistics A, December 2024/25 Solutions

Maps to CS1 Actuarial Statistics exemption.
Weighting within CS1 exemption: 25%.

1. *Similar to examples, first part seen second unseen*
CS1 syllabus objectives 5.1.1

(a) Let R denote the event that the participant recovers, let P denote the event the participant is given the placebo, A be the event they are given drug A and B be the event they are given drug B. Then the probability the participant recovers is given by

$$\begin{aligned}\mathbb{P}(R) &= \mathbb{P}(R|P)\mathbb{P}(P) + \mathbb{P}(R|A)\mathbb{P}(A) + \mathbb{P}(R|B)\mathbb{P}(B) \\ &= (0.1)(0.2) + (0.6)(0.4) + (0.7)(0.4) \\ &= 0.54. \qquad \text{[1 mark]}\end{aligned}$$

Now using Bayes' rule

$$\begin{aligned}\mathbb{P}(A|R) &= \frac{\mathbb{P}(R|A)\mathbb{P}(A)}{\mathbb{P}(R)} \qquad \text{[1 mark]} \\ &= \frac{(0.6)(0.4)}{0.54} \\ &= 0.4444 \qquad \text{[1 mark].}\end{aligned}$$

(b) To determine if the probability of the event, H, of needing hospital care we use chain rule

$$\mathbb{P}(H) = \mathbb{P}(H|P \cap D)\mathbb{P}(D|P)\mathbb{P}(P) + \mathbb{P}(H|P \cap A)\mathbb{P}(D|A)\mathbb{P}(A) + \mathbb{P}(H|B \cap D)\mathbb{P}(D|B)\mathbb{P}(B)$$

where D denotes the event that the participant deteriorates. Therefore

$$\mathbb{P}(H) = (0.3)(0.1)(0.2) + (0.3)(0.1)(0.4) + (0.3)(0.25)(0.4) = 0.048$$
[1 mark]

We also need to calculate

$$\begin{aligned}\mathbb{P}(B) &= 0.4 \\ \mathbb{P}(H \cap B) &= (0.4)(0.25)(0.3) = 0.03 \\ \mathbb{P}(H)\mathbb{P}(B) &= 0.0192. \qquad \text{[1 mark]}\end{aligned}$$

Since $\mathbb{P}(H \cap B) \neq \mathbb{P}(H)\mathbb{P}(B)$ it must be the case that H and B are not independent. [1 mark]

[Total 6 marks]

2. *Similar examples to all parts seen,*
CS1 syllabus objectives 1.1.1, 1.1.3

a) i.

$$\begin{aligned}\mathbb{E}(X) &= \sum_{x=1}^{9} x \cdot p_X(x) \\ &= 1\left(\frac{1}{25}\right) + 2\left(\frac{2}{25}\right) + 3\left(\frac{3}{25}\right) + 4\left(\frac{4}{25}\right) + 5\left(\frac{5}{25}\right) + \\ &\quad 6\left(\frac{10-6}{25}\right) + 7\left(\frac{10-7}{25}\right) + 8\left(\frac{10-8}{25}\right) + 9\left(\frac{10-9}{25}\right) \\ &= \frac{1}{25} + \frac{4}{25} + \frac{9}{25} + \frac{16}{25} + \frac{25}{25} + \frac{24}{25} + \frac{21}{25} + \frac{16}{25} + \frac{9}{25} \\ &= \frac{125}{25} = 5\end{aligned}$$

[2 marks]

ii.

$$\begin{aligned}\mathbb{P}(X < 9) &= 1 - \mathbb{P}(X \geq 9) = 1 - \mathbb{P}(X = 9) \\ &= 1 - \left[\frac{10-9}{25}\right] = 1 - \frac{1}{25} \\ &= \frac{24}{25} = 0.96\end{aligned}$$

[2 marks]

b) For Poisson distribution, the probability mass function is given by:

$$p_Y(y) = \frac{\lambda^y e^{-\lambda}}{y!},\ y = 0, 1, 2, \ldots$$

If $\mathbb{P}(Y = 2) = \mathbb{P}(Y = 3)$ then we can find λ as:

$$\begin{aligned}\frac{\lambda^3 e^{-\lambda}}{3!} &= \frac{\lambda^2 e^{-\lambda}}{2!} \\ \frac{\lambda^3 e^{-\lambda}}{\lambda^2 e^{-\lambda}} &= \frac{3!}{2!} \\ \lambda &= 3\end{aligned}$$

i. $\mathbb{V}ar(Y) = \lambda = 3$ [2 marks]

ii.

$$\begin{aligned}\mathbb{P}(Y > 3) &= 1 - \mathbb{P}(Y \leq 3) \\ &= 1 - [\mathbb{P}(Y = 0) + \mathbb{P}(Y = 1) + \mathbb{P}(Y = 2) + \mathbb{P}(Y = 3)] \\ &= 1 - \left[\frac{3^0 e^{-3}}{0!} + \frac{3^1 e^{-3}}{1!} + \frac{3^2 e^{-3}}{2!} + \frac{3^3 e^{-3}}{3!}\right] \\ &= 1 - 13e^{-3} \\ &\approx 0.3526\end{aligned}$$

[2 marks]

iii. Given that the mean $=2\lambda = 6$, then

$$\begin{aligned}\mathbb{P}(Y > 3) &= 1 - \mathbb{P}(Y \leq 3) \\ &= 1 - [\mathbb{P}(Y = 0) + \mathbb{P}(Y = 1) + \mathbb{P}(Y = 2) + \mathbb{P}(Y = 3)] \\ &= 1 - \left[\frac{6^0 e^{-6}}{0!} + \frac{6^1 e^{-6}}{1!} + \frac{6^2 e^{-6}}{2!} + \frac{6^3 e^{-6}}{3!}\right] = 1 - 61e^{-6} \\ &= 0.8489\end{aligned}$$

[1 mark]

Comment: The new probability $P(Y > 3 \mid 2\lambda)$ is not simply double the original probability $P(Y > 3 \mid \lambda)$. Instead, it reflects a new distribution based on the increased mean. Therefore, doubling the mean λ does not result in doubling the probability $P(Y > 3)$. The relationship between the mean of a Poisson distribution and its probabilities is non-linear; changes in the mean lead to non-linear changes in probabilities.

[1 mark for a sensible comment]

[Total 10 marks]

3. *Similar examples to all parts seen.*
CS1 syllabus objective 1.1.2

(a) Since $\int_0^1 f_X(x)dx = 1$, we have that

$$\int_0^1 (3ax^2 + 2bx + 1)dx = 1$$

$$\left[ax^3 + bx^2 + x\right]_0^1 = 1$$

$$a + b + 1 = 1.$$

Rearranging the equation gives $b = -a$.

[2 marks]

(b) Using the result from part (a), when $a = 5$, the density function becomes $f_X(x) = 15x^2 - 10x + 1$.
Since $f_X(0.5) = -0.25 < 0$, $f_X(x)$ cannot be a density function since a density function must satisfy $f_X(x) \geq 0 \ \forall x$.

[1 mark]

Other equivalent solutions (graphing / identifying that the vertex is negative /showing that the discriminant for the quadratic equation $15x^2 - 10x + 1 = 0$ is positive) are acceptable.

(c) (i) When $a = 1$, the density function is $f_X(x) = 3x^2 - 2x + 1$. We know that $F_X(x) = 0$ for $x < 0$, and $F_X(x) = 1$ for $x > 1$.

When $0 \leq x \leq 1$, we have

$$\begin{aligned} F_X(x) &= \mathbb{P}(X \leq x) \\ &= \int_0^x (3t^2 - 2t + 1)dt \\ &= x^3 - x^2 + x \end{aligned}$$

[2 marks]

(ii) Identify the formula

$$\mathbb{P}(0.4 < X \leq 0.6 \mid X \leq 0.6) = \frac{\mathbb{P}(0.4 < X \leq 0.6)}{\mathbb{P}(X \leq 0.6)}.$$

Calculate

$$\mathbb{P}(0.4 < X \leq 0.6) = F_X(0.6) - F_X(0.4) = 0.456 - 0.304 = 0.152.$$

and

$$\mathbb{P}(X \leq 0.6) = F_X(0.6) = 0.456$$

[1 mark]

Hence,

$$\mathbb{P}(0.4 < X \leq 0.6 \mid X \leq 0.6) = \frac{0.152}{0.456} = \frac{1}{3} = 0.\dot{3}$$

[1 mark]

(iii) Given $Y = e^X$, $0 \leq X \leq 1$, we identify that $\text{Im}Y = [1, e]$. [1 mark]

We know that $F_Y(y) = 0$ for $y < 1$, and $F_Y(y) = 1$ for $y > e$.

When $1 \leq y \leq e$, we have

$$\begin{aligned} F_Y(y) &= \mathbb{P}(Y \leq y) \\ &= \mathbb{P}(e^X \leq y) \\ &= \mathbb{P}(X \leq \ln y) \\ &= (\ln y)^3 - (\ln y)^2 + \ln y \end{aligned}$$

[2 marks]

[Total 10 marks]

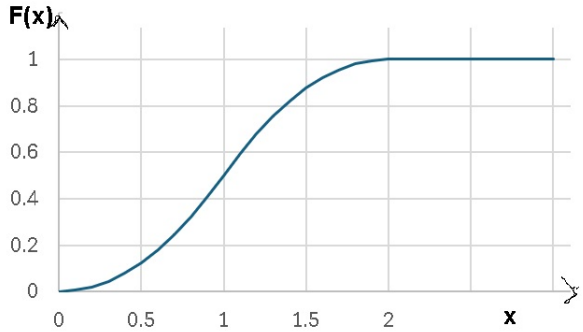
4. *Similar examples to all parts seen.*
CS1 syllabus objectives 1.1.2, 1.1.3

a) The image of X is given by $ImX = [0, 2)$ [1 mark]

In order to plot $F_X(x)$ we need to check whether F_X is continuous, to do this we compare the values at the boundaries of the piecewise regions.

$$\lim_{x\uparrow 0} F_X(x) = 0 = F_X(0),$$
$$\lim_{x\uparrow 1} F_X(x) = \frac{1}{2} = F_X(1),$$
$$\lim_{x\uparrow 2} F_X(x) = 1 = F_X(2). \qquad \text{[1 mark]}$$

The graph of $F_X(x)$ is given by:



[2 marks]

b)

$$\begin{aligned}\mathbb{P}(0.5 < X \leq 2) &= F(2) - F(0.5) \\ &= 1 - \left(\frac{0.5^2}{2}\right) \\ &= 1 - \frac{1}{8} = 0.875\end{aligned}$$

[2 marks]

c)

$$f_X(x) = \frac{dF_X(x)}{dx} = \begin{cases} x, & 0 \leq x < 1 \\ -x+2, & 1 \leq x < 2 \\ 0, & \text{otherwise} \end{cases}$$

[3 marks]

[Total 9 marks]

5. *Similar examples to all parts seen.*
CS1 syllabus objectives 1.2.2, 1.2.4, 1.3.1

(a) Since $ImX = (0,1)$. For $x \leq 0$, and $x \geq 1$, $f_X(x) = 0$. For $0 < x < 1$, the probability density function $f_X(x)$ of X is

$$\begin{aligned} f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)dy &= \int_x^1 8xydy \\ &= 8x\left[\frac{y^2}{2}\right]_x^1 \\ &= 8x\left(\frac{1-x^2}{2}\right) \\ &= 4x(1-x^2) \end{aligned}$$

$$f_X(x) = \begin{cases} 4x(1-x^2) & ,0 < x < 1, \\ 0 & ,\text{otherwise.} \end{cases}$$

[2 marks]

Since $ImY = (0,1)$. For $y \leq 0$, and $y \geq 1$, $f_Y(y) = 0$. For $0 < y < 1$, the probability density function $f_Y(y)$ of Y is

$$\begin{aligned} f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)dy &= \int_0^y 8xydy \\ &= 8y\left[\frac{x^2}{2}\right]_0^y \\ &= 8y\left(\frac{y^2}{2}\right) \\ &= 4y^3 \end{aligned}$$

$$f_Y(y) = \begin{cases} 4y^3, & 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

[2 marks]

(b) From the marginal density of X we have

$$\begin{aligned} \mathbb{E}[X] &= \int_{-\infty}^{\infty} x f_X(x)dx \\ &= 4\int_0^1 x^2(1-x^2)dx \\ &= 4\int_0^1 x^2 - x^4 dx \\ &= 4\left[\frac{1}{3}x^3 - \frac{1}{5}x^5\right]_{x=0}^1 \\ &= 4\left(\frac{1}{3} - \frac{1}{5}\right) = \frac{8}{15}. \end{aligned}$$

[1 marks]

From the marginal density of Y we have

$$\begin{aligned}\mathbb{E}[Y] &= \int_{-\infty}^{\infty} y f_Y(y) dy \\ &= 4\int_0^1 y^4 dy \\ &= 4\left[\frac{1}{5}y^5\right]_{y=0}^1 \\ &= \frac{4}{5}.\end{aligned}$$

[1 marks]

(c) For $0 < x < y < 1$, the conditional density $f_{Y|X}(y|x)$ is given by

$$\begin{aligned}f_{Y|X}(y|x) &= \frac{f_{XY}(x,y)}{f_X(x)} \\ &= \frac{8xy}{4x(1-x^2)} \\ &= \frac{2y}{(1-x^2)}\end{aligned}$$

$$f_{Y|X}(y|x) = \begin{cases} \dfrac{2y}{(1-x^2)}, & 0 < x < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

[2 marks]

(d) When $x = \frac{1}{2}$,

$$f_{Y|X}(y|x) = \begin{cases} \frac{8y}{3}, & \frac{1}{2} < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

[1 mark]

$$\begin{aligned}E(Y|X = \frac{1}{2}) = \int_{\frac{1}{2}}^1 y \cdot f_{Y|X}(y|\frac{1}{2})dy &= \int_{\frac{1}{2}}^1 \frac{8}{3}y^2 dy \\ &= \frac{8}{9}\left[y^3\right]_{\frac{1}{2}}^1 = \frac{8}{9}(1 - \frac{1}{8}) = \frac{7}{9}\end{aligned}$$

[2 marks]

(e) We can calculate the covariance using

$$\mathbb{C}\text{ov}(X,Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y).$$

We first calculate $\mathbb{E}(XY)$:

$$\begin{aligned}\mathbb{E}(XY) &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy f_{X,Y}(x,y)dxdy \\ &= 8\int_0^1\int_0^y x^2y^2 dxdy \\ &= 8\int_0^1 y^2\left[\frac{1}{3}x^3\right]_{x=0}^{x=y} dy \\ &= \frac{8}{3}\int_0^1 y^5 dy \\ &= \frac{8}{3}\left[\frac{1}{6}y^6\right]_{y=0}^{y=1} \\ &= \frac{4}{9}.\end{aligned}$$

[2 marks]

Now using that $\mathbb{E}[X] = 8/15$, $\mathbb{E}[Y] = 4/5$ we have

$$\mathbb{C}\text{ov}(X,Y) = \frac{4}{9} - \frac{8}{15}\frac{4}{5} = \frac{4}{255}.$$

[1 mark]

(f) Since $\mathbb{C}\text{ov}(X,Y) = \frac{4}{225}$ is a positive value, it indicates a positive linear relationship between X and Y. This means that as X increases, Y tends to increase slightly as well. [1 mark for any sensible comment]

[Total 15 Marks]

6. *All parts unseen.*
CS1 syllabus objective 1.2.6. 1.5.1

(a) Here $T_i \sim \text{Exp}(1)$ so $\mathbb{E}(T_i) = 1$ and $\mathbb{V}\text{ar}(T_i) = 1$. [1 mark]
Let $X_i = cT_i - 85$ then $\mathbb{E}(X_i) = c - 85$ and $\mathbb{V}\text{ar}(X_i) = c^2$. Hence $\mathbb{E}(P_n) = n\mathbb{E}(X_i) = n(c-85)$.

[1 mark]

Since the X_i are independent we have

$$\mathbb{V}\text{ar}(P_n) = \sum_{i=1}^{n} \mathbb{V}\text{ar}(X_i) = nc^2$$

[1 mark]

(b) By the central limit theorem

$$\frac{(P_n - n\mathbb{E}(X_i))}{\sqrt{n\mathbb{V}\text{ar}(X_i)}} \sim \text{N}(0,1).$$

[1 mark]

Substituting in our values of $\mathbb{E}(X_i)$ and $\mathbb{V}\text{ar}(X_i)$ and rearranging gives

$$\frac{(P_n - \mathbb{E}(P_n))}{\sqrt{n}} \sim \mathrm{N}\left(0, c^2\right).$$

[1 mark]

(c) Determine the value of c which corresponds to a fair price, i.e. such that $\mathbb{E}[P_n] = 0$ for any n. The $\mathbb{E}[P_n] = n(c-85)$ so setting $c = 85$ gives a fair price.

[1 mark]

(d) Suppose that the company sets $c = 90$, use the central limit theorem to find a function $\ell(n)$ such that

$$\lim_{n\to\infty} \mathbb{P}(\ell(n) \leq P_n) = 0.95.$$

Let

$$Z = \frac{P_n - n(c-85)}{\sqrt{nc^2}}$$

then $Z \sim \mathrm{N}(0,1)$. From the normal tables we have

$$\mathbb{P}(-1.6449 \leq Z) = 0.95.$$

[1 mark]

Substituting for Z and rearranging

$$\mathbb{P}(n(c-85) - 1.6449\sqrt{n}c \leq P_n) = 0.95$$

[2 marks]

Finally setting $c = 90$ gives the functions

$$\ell(n) = 5n - (1.6449)(90)\sqrt{n} = 5n - 148.041\sqrt{n}.$$

[1 mark]

[Total 10 marks]